



A framework to study requirements-driven collaboration among agile teams: Findings from two case studies



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ABSTRACT

Stakeholders' collaboration is required to develop requirements in agile software development. Requirements engineering and agile methods share common grounds as they both focus on stakeholder collaboration. The key issue is finding a way to study collaboration driven by requirements in geographically distributed agile teams. In this paper, we aim to propose a framework to study collaboration driven by requirements among agile teams and determine the impact of their collaboration patterns on the iteration performance. We define collaboration in terms of communication as *information exchange among members*, and awareness *knowledge of others*. Two case studies were conducted to examine communication and awareness network patterns among distributed agile teams. Data were collected through questionnaires, semi-structured interviews and onsite observation. The findings revealed that the framework aids in determining the core members, collaboration trends, clustering tendency, communication and awareness reciprocity of the teams, small worldliness and centralisation behaviour of the networks and iteration performance of the agile teams. The framework has implications for the industrial practitioners, i.e. managers to learn about their team's collaboration in order to take measures for performance improvement. At the same time, the researchers can use this framework to study other social aspects in variable settings to produce more empirical results.

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1. Introduction

Requirements engineering (RE) is part of a human-centric social process, called software engineering. The requirements are treated as a unit of work for which the stakeholders collaborate, i.e. communicate with each other, share knowledge and exchange information. The collaboration driven by the requirements during their development and management to the downstream artefacts, i.e. code, design, etc., is defined as requirements-driven collaboration (RDC) (Damian, Kwan, & Marczak, 2010).

Agile methods, being collaborative in nature, entail an organic management of requirements, unlike traditional software development methods (Cao & Ramesh, 2008); they also serve as an alternative approach as they address the challenges posed by traditional software development (i.e. communication gaps, over-scoping (Bjarnason, Wnuk, & Regnell, 2010, 2011)). Agile methods emphasise extensive collaboration between customers and developers, and encourage small, self-organised and collocated teams (Sharp, Robinson, & Hall, 2003). In such a dynamic software

development process, requirements are highly volatile and constant collaboration is essential to cope with the ever-changing requirements for risk mitigation due to dependencies (Martakis & Daneva, 2013). Developer collaboration is dependent on the communication of changes and of new tasks, as well as on the awareness of what others are doing, whether they are available to help, or what they know that can help one's work (Damian, Izquierdo, & Singer, 2007). Agile methods stress stakeholders' collaboration throughout the development process. Unlike traditional software development teams, agile teams are cross-functional, closely knit and highly interactive. The information is believed to be highly dispersed in the agile teams due to members working closely and changing roles constantly. Therefore, this difference in team structures raises interesting aspects to be explored.

In the literature, there are just a handful of empirical research studies that focused on the social aspects of agile requirements engineering, especially communication (e.g. Cataldo & Ehrlich, 2011; Iivari & Iivari, 2011; Mishra & Mishra, 2009; Pikkarainen, Haikara, Salo, Abrahamsson, & Still, 2008)). It is mentioned in a study carried out by Komi-Sirviö and Tihinen (2005) that 74% of the problems encountered by distributed development teams are related to communication due to cultural differences (Summers, 2008), language barriers (Kajko-Mattsson, Azizyan, & Magarian,

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2010), etc. Therefore, communication among agile has been studied more widely as compared with the other socio-technical aspects in agile teams, e.g. trust, knowledge sharing (Licorish & MacDonell, 2014). Moreover, based on our previous research, agile practitioners also declared communication and awareness as the most relevant socio-technical aspects of collaboration driven by requirements (Inayat, Salim, Marczak, & Kasirun, 2014). Therefore, the proposed framework is an initiative to provide a formalised set of steps to study the socio-technical aspects (i.e. communication and awareness) among agile teams empirically supported by two real-world case studies.

The framework intends to close the research gap by studying the characteristics of the collaboration patterns of agile teams. Moreover, the available literature remained silent on several issues such as team performance in variable development atmospheres involving agile methods with a high rate of change acceptability. Hence, this justifies the need for proposing a framework for agile teams with high collaboration, requirements volatility and small teams. A formal set of steps was introduced to study the collaboration among agile teams in terms of their communication and knowledge of each other. The purpose of this framework is to investigate the relevant aspects of RDC for agile teams, to further the knowledge on the topic and to find the impact of RDC on iteration performance.

2. Related work and motivation

Agile methods focus on team collaboration through frequent face-to-face communication (Beck, 2000). However, only a handful of empirical researches explored social aspects of collaboration among agile teams, including communication (Cataldo & Ehrlich, 2011; Pikkarainen et al., 2008), trust (McHugh, Conboy, & Lang, 2012; Tjørnehøj, 2012), organisational culture (Robinson & Sharp, 2005; Susan, 2011) among others. Some of the studies that focused on communication are discussed in this paper.

Abdullah, Honiden, Sharp, Nuseibeh, and Notkin (2011) studied communication patterns of agile teams while carrying out requirements engineering activities (i.e. elicitation, analysis, modelling, etc.). The results revealed that the concept of shared conceptualisation among agile teams supports the requirements engineering activities by introducing common goals.

Cataldo and Ehrlich (2011) studied communication patterns of distributed agile teams to find out the impact of hierarchical and small world network structures on iteration performance and outcome quality. The results revealed that hierarchical network structures have a strong positive effect on performance and negative effect on outcome quality. However, small world structures showed strong positive effect on quality and negative effect on performance.

Pikkarainen et al. (2008) investigated the effects of variable agile methods, i.e. XP and Scrum on agile teams' communication. The authors proposed a theoretical framework to study the impacts of agile methods on communication among the team members. The results revealed that agile methods improve formal and informal communication among team members. Informal communication is defined as the communication that happens outside the official reporting hierarchy of the team (Herbsleb & Grinter, 1999). However, the involvement of a large number of stakeholders and use of different communication mediums can cause problems.

The above discussion shows that the collaboration patterns of agile teams for requirements engineering activities yet need to be explored thoroughly. Although there are a few literature studies available on social factors involved in agile software development methods, agile way of dealing with requirements still warrants further research. In addition, there is a need of a systematic approach to guide the investigation of social factors in agile teams. In terms

of research on collaboration driven by requirements engineering activities in traditional software development teams, a set of steps based on social network analysis (SNA) was proposed by Damian et al. (2010). However, Damian et al. (2010) remained silent on the issues like studying the effects of collaboration on team performance. Therefore, the proposed framework is an initiative to close this research gap and provide a formalised set of steps to study socio-technical aspects (i.e. communication and awareness) among agile teams.

3. Data sources for the formulation of framework

The two data sources, which are systematic literature reviews (SLR1 and SLR2) and survey are used as a basis for the framework formulation, which are described in this section and depicted in Fig. 1.

3.1. Online survey

The survey study was conducted to identify the most relevant socio-technical aspects from the perception of industry practitioners using agile methods (Inayat, Salim, Marczak, & Kasirun, 2014). The main aims were to: (i) find out the perception of agile practitioners about collaboration; and (ii) identify the socio-technical aspects which agile teams find relevant while collaborating with each other for certain interdependent requirements (user stories). We used an online questionnaire for gathering responses from industry practitioners. A total of 103 responses were collected from all over the world. About three-quarters of the responses were gathered through the professional virtual community (PVC), i.e. [linkedin.com](https://www.linkedin.com) while about one-quarter was collected through e-mail (50 email invites were sent and 30 people replied, considered 26). The survey results showed that practitioners perceived collaboration mainly as communication among peers, knowledge of other team members (awareness) and trust in each other. Moreover, the practitioners selected communication, awareness and knowledge sharing as the most relevant aspects of collaboration among teams. Therefore, communication and awareness were chosen to be studied in detail for agile teams.

3.2. Systematic literature review

Two systematic literature reviews (SLRs) were conducted:

- SLR1 (Inayat, Salim, Marczak, Daneva, Shamshirban, 2015) is conducted to further the understanding of the underlying concepts of agile RE, i.e. agile teams, user stories, agile RE practices and challenges.
- SLR2 is conducted to review communication and awareness in the literature in order to further the understanding of both aspects in RE activities.

These reviews were conducted according to the available guidelines for conducting systematic literature reviews (Kitchenham & Charters, 2007) to develop further understanding of the topic. Of the 543 initial papers located in well-known electronic research databases, 21 relevant papers were extracted for SLR1. Likewise, 16 studies were selected out of 340 studies for SLR2.

The findings of SLR1 (Inayat, Salim, Marczak, Daneva, Shamshirban, 2015) revealed the agile RE practices, agile RE challenges and implications. The findings suggest that agile requirements engineering in a research context needs additional attention and more empirical results are required to better understand the impact of agile requirements engineering on team performance and outcome quality. The implications stated that the

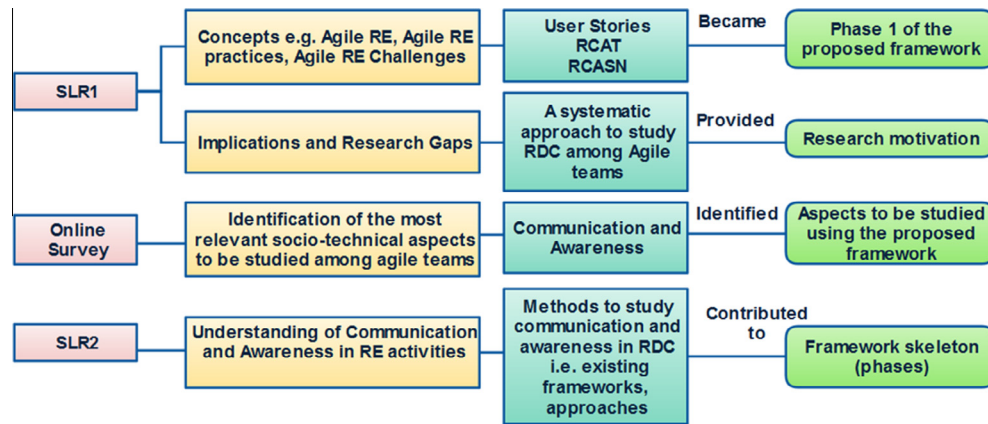


Fig. 1. Research activities carried out during the research process of framework formulation.

empirical results generated from communication patterns and collaboration studies among agile teams should be utilised by project managers to orient their teams towards the implementation of certain collaboration structures, i.e. hierarchal or centralised among them for maximum performance. Therefore, focus on studying RDC among agile teams in detail and providing more empirical findings to strengthen the knowledge in this area was recommended.

The results of SLR2 have, firstly, helped to gather more information on the two identified socio-technical aspects. Secondly, the results have contributed to defining the methods and approaches used in the literature in order to conduct further study on them. This helped in forming the basic structure of the proposed framework. The use of SNA for studying the social aspects of RDC encouraged the usage of SNA measures for studying communication and awareness for agile teams in this study.

The theoretical integration of the data gathered from the data sources discussed in Sections 3.1 and 3.2 was performed. The findings from both of the data sources are combined to thoroughly analyse the new and overlapping concepts put forward by the survey and systematic literature reviews, and to aggregate and relate the existing theories in literature, as shown in Fig. 2.

The framework is then practically implemented by carrying out two case studies with distributed agile teams. The results further the knowledge of the topic and provide implications for the researchers and industry practitioners.

4. A framework to study requirements-driven collaboration among agile teams

Fig. 3 shows the proposed framework. The framework consists of four phases that guide the investigation of communication and awareness among agile teams.

Phase 1 is “Conceptualisation”, in which the underlying concepts such as requirements centric agile teams and requirements centric agile social networks are defined. Phase 1 provides the conceptual basis for the proposed framework.

Phase 2 is “Visualisation”, in which communication and awareness networks of RDC among agile teams is visualised in the form of networks called sociograms. The sociograms help to define certain characteristics of the communication and awareness networks, i.e. members involved in networks, size of the networks, and density of the networks. In addition, it also helps to estimate the communication and awareness among the teams.

Phase 3 is “Characterisation”, in which the structures of requirements-centric communication and awareness patterns among agile teams are studied to determine teams’ centralisation,

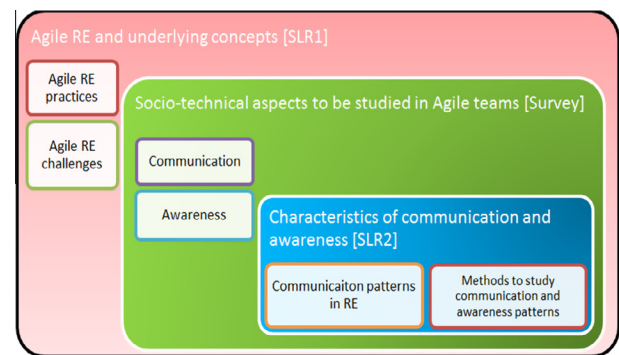


Fig. 2. Concepts gathered from data sources.

reciprocity, clustering, etc. In addition, characteristics of the information flow patterns of the agile teams are studied to find out the central and pivotal members of the teams.

Phase 4 is “Impact analysis”, in which the impact of communication and awareness networks on agile teams’ iteration performance is measured. The detailed description of each phase is as follows.

4.1. Phase 1 – Conceptualisation

The following are the concepts defined to study RDC among agile teams.

- **Requirements centric agile team:** Agile teams comprise self-managing and self-organising groups of people, who make decisions themselves and coordinate with each other to jointly develop the tasks (Schwaber & Beedle, 2001). Drawing on the work of Damian et al. (2010), we proposed the concept of a Requirements Centric Agile Team (RCAT) (Inayat, Marczak, & Salim, 2013). The RCAT refers to the group of people collaborating around a certain set of interdependent user stories (requirements) broken down into tasks as well as on downstream artefacts, i.e. design, code, etc. This group of members includes anyone who is assigned to work on the project, including for example the participation of the customer (customer representative or proxy customer). It also includes anyone who “emerges” throughout the life cycle for some reason, i.e. additional people who collaborate at any given moment. These people are called “emergent members” (Inayat et al., 2013).
- **Requirements centric agile social networks:** Requirements Centric Social Networks (RCSN) is defined as a network of a group of people working on certain interdependent requirements and

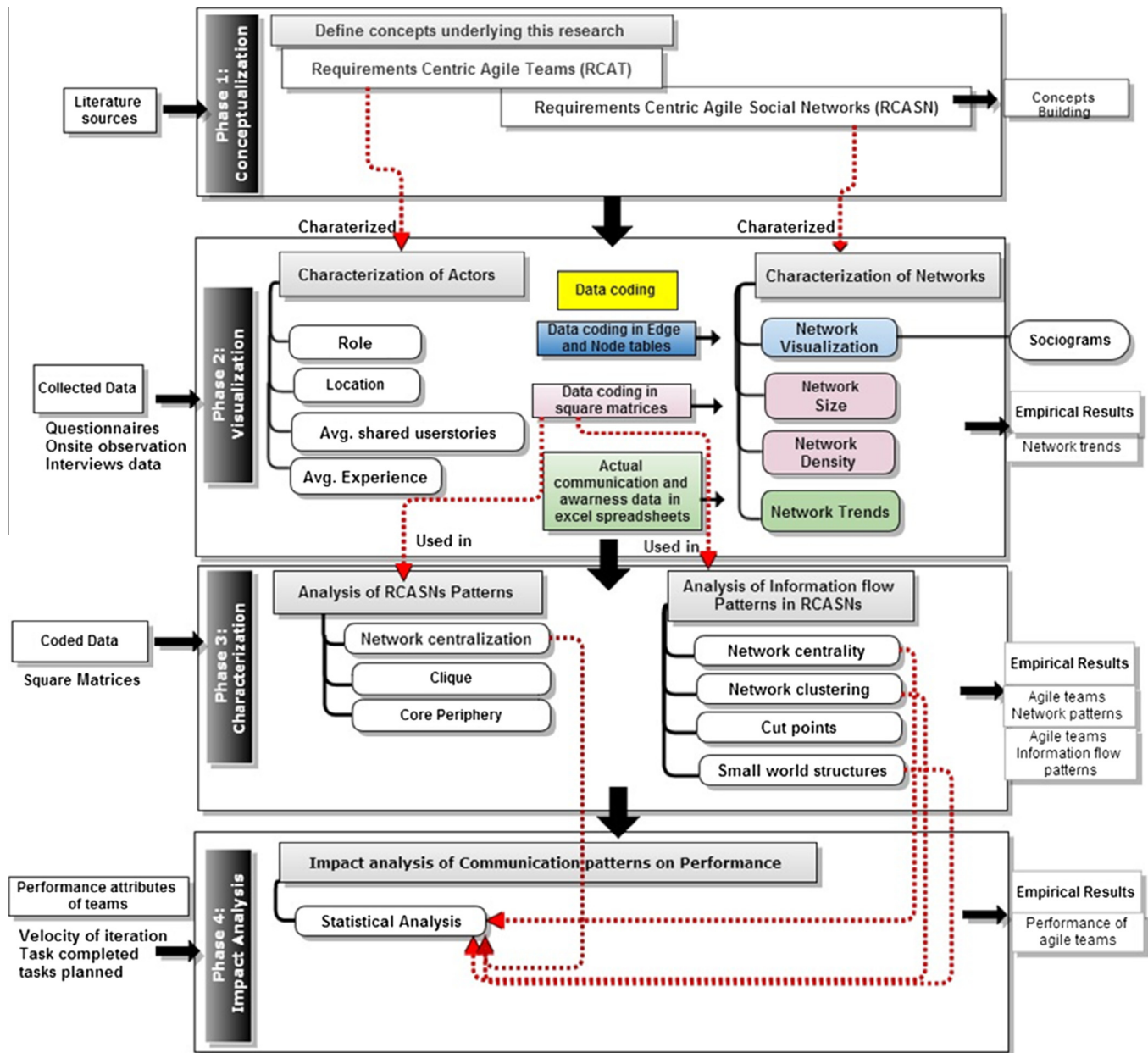


Fig. 3. Framework for studying communication and awareness among agile teams.

their downstream artefacts (Damian et al., 2010). Similarly, we define such a network for an RCAT as a group of cross-functional and self-organising agile team members, actors and their inter-relationships, ties. Such network is defined as a Requirements Centric Agile Social Network (RCASN) (Inayat et al., 2013). A tie in a certain RCASN represents a team member's communication regarding user stories or team member's awareness about interrelated user stories. The RCASN(s) created for RCAT(s) are based on questions such as "Who communicated for a certain user story with whom?" "Who was informed about any change in a certain user story?" and "Who else needs to be contacted for a certain user story?".

4.2. Phase 2 – Visualisation

The collaboration among agile teams was visualised from the lens of actors, networks and their characteristics, as described below.

4.2.1. Characterisation of actors

Who are involved in the communication and awareness networks? The characterisation of actors is performed to find out who are involved in the networks. It involves finding out the important details concerning demographics and project such as experience level, geographical location, and average shared user stories. For instance, role of emergent members in networks and role of experienced members in networks can be of help in determining their collaboration patterns.

4.2.2. Characterisation of networks

The characterisation of networks is performed through visualisation, measuring size, density and analysing their trends.

- *Network visualisation:* To answer "How to visualise the ongoing communication and awareness in agile teams?" the network visualisation phase is introduced. The networks are visualised in the form of sociograms. The sociograms present actors (team

members) as “nodes” and their relationship (communication or awareness) as “ties”. The visual examination of the sociograms helps in determining network size, density, and trends. The sociograms of the networks are created by encoding the communication and awareness data gathered through questionnaires. Node and edge tables are created for each network and then uploaded to the network visualisation tool known as Gephi (www.gephi.com). Node tables comprised information, i.e. node name, node label, and node id. Edge table consisted of source, target, weight (frequency of communication) and direction of the edge (directed or undirected).

- **Network size:** To answer “How many team members are involved in the communication and awareness networks?” the network size is recorded. The size determines the number of nodes or actors involved in each network. It helps in determining the number of actors who participated in certain kind of discussion. For instance, if size of the network presented is 11, this means that 11 members communicated for certain kind of discussion. It helps to determine: (i) the number of members involved; (ii) the participation trends of members, like most of the developers communicated for code issues discussions; (iii) the amount of collaboration required for a certain kind of requirement and communication reasons.
- **Network density:** To answer “How dense are the communication and awareness network?” density measure is applied. Density is defined as the proportion of all of the existent ties (Freeman & Mulholland, 1979). It is calculated by dividing the present ties by the total number of possible ties. Density helps to assess how tightly coupled the team is. Low density shows less communication between members than expected. We consider density equal or above 0.5 as a high-density value, meaning that more than half of the possible ties have been reported as present.
- **Network trends:** To answer “How much communication and awareness exist cross and within-roles, cross and within-locations and cross and within-task assignment?” network trends are manually calculated. The network trends determined the analysis of network structure; the number of ties linking the actors helps in finding the collaboration patterns of teams. Each member is examined with respect to three aspects: (i) position in team, (ii) location, and (iii) task assignment. Using the pair wise coding format, datasets were imported to Excel to calculate the ties statistics for all of the communication and awareness networks. For each pair of members with existent link, we considered them as one tie. The communication and awareness were considered directional, i.e. if a pair “source-target” reported communication and this same pair “target-source” also reported communication, two instances of communication were considered. This involved a visual inspection of networks and a manual calculation process for filtering the dataset. Second, the data were recoded into matrix format with a distance matrix (explaining the locations of actors per network) and the correlation between communication and awareness was calculated. This pair-wise coding format helps in determining the actors in pivotal position (with maximum number of ties), actors with maximum inter-role communication and awareness, actors with maximum cross-site communication and awareness. In addition, the correlation helped in understanding the effect of distance on awareness and communication for distributed agile teams.

4.3. Phase 3 – Characterisation

In this section, the network characterisation is explained with respect to agile teams’ patterns and information flow.

4.3.1. Characterisation of network patterns

The characterisation of networks aimed to find answers to these questions: (i) Do agile teams have centralised communication and awareness networks structures? (ii) Do agile teams have members working more closely together compared with others? (iii) Are there lone or isolated team members or groups of members in agile teams? To study the communication and awareness network patterns, a detailed analysis was conducted using SNA measures, as explained below.

- **Network centralisation:** To answer “Do agile teams have centralised communication and awareness network structures?” network centralisation is used. The centralisation value for networks determines the dispersion of ties around nodes (Freeman & Mulholland, 1979). If a network is centralised, the value of centralisation lies closer to 0. On the contrary, if the nodes have dispersed ties with multiple nodes, the network is decentralised; thus, the value of centralisation is less than 1. The centralisation measure gives an idea about the chain of command in a team, e.g. if the centralisation value for a network is high, it indicates that the network is centralised and vice versa.
- **Clique:** To answer “Do agile teams have members working more closely together compared with others?” clique is used. A clique is defined as a subset of a network in which members (nodes) are more closely tied to one another than with the rest of the members of the network (Hanneman & Riddle, 2005; Mchugh, 1990). For instance, in real life social networks, friendship networks contain such cliques formed among members on the basis of common interests, age, gender, ethnicity, etc. Clique is useful in providing inter-team collaboration structure. The cliques help in understanding which roles were closely communicating for which communication reasons and which actors tended to communicate more with each other.
- **Core periphery:** To answer “Are there lone or isolated team members or groups of members in agile teams?” core periphery is used. It describes the network structure in terms of two sets of nodes, i.e. core and periphery. Core has nodes that are closely connected to each other, and periphery has nodes that are loosely connected to each other. Core-periphery measure has been used in the literature for the following purposes: (i) to identify creative actors by determining intermediate positions (Cattani & Ferriani, 2008), and (ii) to measure impact of leadership development on leadership networks (Hoppe & Reinelt, 2010).

4.3.2. Characterisation of information flow among agile teams

The characterisation of information flow patterns aims to answer the following questions: (i) Do agile teams have some actors playing a central part in information flow? (ii) Do agile teams have such member(s) who play a pivotal part in information exchange? (iii) Do agile teams have the tendency to cluster with the increase in communication? (iv) Does the agile team show small world network structure? To identify the patterns of information exchange among agile teams, several SNA measures are used, and are explained below.

- **Degree centrality:** To answer “Do agile teams have some actors playing a central part in information flow?” degree centrality is used. Degree centrality is the number of ties a node has (Freeman & Mulholland, 1979). The number of ties a node emits is “out-degree” and the number of ties it receives is “in degree”. The centrality measure can determine power (Gomez et al., 2003) and activity of actors in a social network (Freeman & Mulholland, 1979). For an agile team, centrality measure helps in determining the actors with the highest degree values and helps in identifying the most active, powerful, and central actors in the networks.

- **Cut point:** To answer “Do agile teams have such member(s) who play a pivotal part in information exchange?” cut point is used. The cut point determines the actors or actors (called cut sets) in the networks who if removed can cause information loss due to lost connections with other nodes. Cut points are used to detect the key players in a social network (Borgatti, 2006) and information brokers in traditional software development teams (Damian et al., 2010).
- **Clustering coefficients:** To answer “Do agile teams have the tendency to cluster with the increase in communication?” clustering coefficients are used. The clustering coefficients are one of the important properties of a graph or network which determines the degree or extent of nodes to stay closer or have better connection with neighbouring nodes.
- **Small world network structure:** To answer “Do agile teams show small worldliness behaviour?” the small world network structures of the teams were observed. Small world phenomena occurs when two nodes are connected to one another through a small sequence of acquaintance nodes in between (Kleinberg, 2000). On the same footings the small network structures are defined as small and highly dense clusters of nodes connected to a few other nodes. In the context of collaboration translated to communication between software development teams, this means that a team member is surrounded by dense networks of other team members, who are all communicating with each other.

4.4. Phase 4 – Impact analysis

Performance of agile teams has been studied in literature with respect to number of defects and rework per iteration (Cataldo & Ehrlich, 2011), time to complete a certain task (Espinosa, Slaughter, Kraut, & Herbsleb, 2007), time to complete rework on requirements changes (Cataldo & Herbsleb, 2008), and quality, division of work, iteration amendments and teams' satisfaction (Drury-Grogan, 2014). In practice, some of the performance metrics used by the industry practitioners to measure agile team's performance are LOC (line of code, customer satisfaction, defect count, lead time, unfinished stories, etc.). However, identifying the agile team's productivity factors in a socio-technical system is an area that is considered to be under-researched and challenging (Melode O, Cruzes, Kon, & Conradi, 2013). Therefore, this study aims to find the impact of agile teams' collaboration, i.e. their mutual communication and awareness of each other's general traits, availability, current task and work status, on iteration performance. We define iteration performance as the number of user stories completed per iteration, number of user stories repeated per iteration (amount of rework), and velocity per iteration.

5. Structural validation

The framework is structurally validated through questionnaire-based semi-structured interviews. The validation results provided confidence in the correctness of the framework constructs to attain the objectives of this study. Based on the respondents' suggestions, the framework is practically demonstrated through a multiple case study performed at two international agile savvy software organisations with distributed teams in Asia, Europe and America.

The objective of the structural validation process is to gather feedback on the constructs of the proposed framework and to make the necessary improvements according to the suggestions. Expert opinion of well-versed academicians and researchers were used for this purpose. The GQM statement to perform this validation is shown in Table 1.

Table 1
GQM statements to perform validation with researchers.

To analyse	The processes to study the socio-technical aspects of RDC in agile teams
In order to evaluate	The constructs of the proposed framework
From the perspective of	Researchers in the field of requirements engineering and agile software development methods
In the context of Because	Face-to-face and online questionnaire based interview The suggestions of researchers and academicians help to improve and validate the results

A semi-structured interview strategy was chosen to obtain information from the experts. The interview technique was intentionally chosen over others, such as questionnaire-based survey, so as to gain more of the expert's opinion, which would not have been possible in the case of a Likert scale questionnaire. Moreover, interactive sessions can lead to more constructive feedback, which is impossible to gain through an open-ended questionnaire. However, this technique is time consuming right from its preparation to sending invites and waiting for responses, and then having interviews conducted at the agreed time for each of the experts. Nevertheless, the feedback obtained was useful and led to slight modifications to the proposed framework, making it more applicable for the industry.

5.1. Participants

The participants of this questionnaire-based interview were selected on the basis of their professional background, research interests and their publications after determining their due willingness to participate. The invitations, along with the details of the validation study, were sent to 15 experts within the field. However, only six experts were willing to participate. The research interests of the participants were in requirements engineering, agile software development methods, and social aspects in requirements engineering.

5.2. Research instrument and procedure

A questionnaire was prepared (shown in Appendix A, Table A.1) and used to conduct the interviews. The questions were not asked in the same order as they were written in the questionnaire. A copy of the questionnaire was also sent to the participants to avoid any kind of miscommunication. The interviews were conducted online through Skype chat service while face-to-face interviews were conducted with the accessible participants. A handout comprising the necessary information to be shared was prepared and sent beforehand to the participants. In addition, the interviews were followed by a short 10–15 min oral presentation to avoid any ambiguity and for a clear explanation of the results to gain concise and appropriate feedback. The interviews were semi-structured and the sequence of questions was changed according to the direction of the discussion. The interviews were recorded and field notes were taken so as to avoid any missing information. Later, while transcribing the interviews, the field notes were matched and linked to the discussion. Post interview emails were also sent to the experts to clarify certain aspects.

5.3. Analysis and results

First, the interview recordings were transcribed carefully by replaying them at least twice to avoid any missing information. Each of the transcribed files is saved separately with the

Table 2

Categories and their descriptions from the transcribed interviews.

Category	Sub category	Description	Interviewee
Feasibility	Suggestion Agreement	Application in industrial environment will add to it	B
		A feasible approach for studying collaboration	A
		An approach that provides insights on the teams' collaboration	E
		It is feasible to study collaboration of agile teams	F
		It can be followed by studying other social aspects in the future by researchers	D
SNA and statistical measures	Disagreement	Processing should be made automatic for managers	B
	Suggestions	More SNA measures can be added based on the intention of the manager and situation of the team	D
		Some SNA measure that determines the position of a particular node in the network can add to the analysis	C
	Agreement	Adequate SNA measures for network analysis	B
		Enough SNA and statistics for finding detailed insights on a team's collaboration attributes	A
		Covers almost every aspect of team's collaboration	D
Improvement	Disagreement	Applying all of them would be too much for a Manager to do	C
Improvement	Suggestions	Deploy in industry and gain feedback on the results	C
		Deployment in industry should be made easier	E

Table 3

Concerns and necessary actions taken to mitigate them.

Concerns	Action taken
Application in industrial environment will add to it	The framework is implemented through two real world case studies explained in Section 6
Processing should be made automatic for the managers	A prototype automating the framework was planned based on this feedback
More SNA measures can be added based on the intention of the manager and situation of the team	The SNA measures were added after industry validation based on the practitioners' explanation of the situations of their interest
Some SNA measure that determines the position of a particular node in the network can add to the analysis	Another phase of node characterisation can be added to the framework
Deploy in industry and gain feedback on the results	The framework is deployed through two real world case studies explained in Section 6
Deployment in industry should be made easier	A prototype automating the framework was planned based on this feedback

interviewee's identical code allotted for privacy purpose, i.e. A, B, C, etc. Hence, there were a total of 6 transcribed files from A to F. Then, the coding process was started. During the transcription process, a few minor ideas on the categories of comments emerged. From here, the main categories were later on formulated such as feasibility, measures, and steps. These main categories were looked into in the transcribed files for another three sub-categories, which are suggestion, disagreement, and agreement. The descriptions of these sub categories are such that (i) suggestion is any action required for the improvement or change in the proposed framework, (ii) disagreement is any concern of the experts on the current proposed framework, and (iii) agreement is the acceptance by the experts of the proposed framework steps and measures with no further action required in terms of any change or deletion. The detailed coding of the experts interviews is shown in Table 2.

Based on the suggestions, agreement and disagreements shown in Table 2, the necessary actions were taken before validating the suitability and applicability of the study by industry practitioners. The list of the necessary actions taken according to the experts' advice is shown in Table 3.

In a nutshell, from the structural validation performed, the proposed framework constructs have been proven to be helpful for studying collaboration and finding its impacts on performance. The experts declared "agreement" on the phases of the proposed framework for investigating collaboration among agile teams. The respondents showed "agreement" that the proposed framework has its applicability in the industry and also declared the framework "helpful" in finding the team's collaboration insights and investigating impacts on team's performance. However, the experts stressed framework automation to provide a ready to use package for the managers to assess the collaboration health of their teams. In addition, node characterisation was also suggested, which can be a potential research area along with the framework automation.

6. Case study implementing the proposed framework

We report the empirical results of two case studies following agile methods by using questionnaires, semi-structured interviews, onsite observation and document analysis. The case study approach was implemented with reference to the steps given by Yin (2009) for a case study research method, i.e. preparing data collection, actual data collection, analysing case study evidence and reporting case studies.

The two projects selected for this research were purposefully chosen to be midsized (6–8 months), having an iteration size not longer than 1 month, following agile methods and its variants, and having distributed teams. We studied two iterations of each project to find the collaboration changes over the period of time. The two projects selected for this research were carried out by distributed teams located in Asia and Europe. The teams followed the Scrum method for carrying out midsized (4–6 months spanned) products followed by several release versions after a year. The requirements originated from external customers (outside the organisation) and were formalised into a feature list by project manager and executives. The feature list was then handed over to the team to make possible user stories and divide among them based on their expertise and previous experience. The distributed teams used virtual collaboration infrastructure like Skype and large touch screens for video conferencing for collaboration.

Project Alpha is a mobile internet security provider software that ensures user's information safety against digital threats on the Web. The team comprised 10 team members distributed in Malaysia (MY) and Finland (F), including: 1 Project Manager (PM-F), 1 Product Owner (PO-F), 5 Software Developers (4 Dev-MY, 1 Dev-F), 2 Quality Engineers (Tester-MY), and 1 User Experience Designer (UX-F).

Project Beta is a web-based physical and electronic data indexing and cataloguing solution that enables one to store, manage,

track, and utilise the information assets ranging from core rock samples to well seismic and map data stored at multiple storage locations. The team consisted of 5 members distributed at two locations, Malaysia (MY) and the United States (US), with the following distribution: 1 Project Manager (PM-US), 3 Software Developers (1Dev-US, 2Dev-MY), and 1 Technical Architect (TA-US).

Both of the teams use the JIRA tool for project management and user story records, Scrum burn down charts, online tracking and charts for iteration progress control.

6.1. Data collection

The data were collected in two phases. In phase 1, data about project (nature, time span, method in use, iterations, iteration size, number of user stories, etc.), team (size, roles, responsibilities of roles and work flow), processes and practices (collaboration practices, collaboration tools, artefacts used and documents involved) were gathered. However, in phase 2, the emphasis was more on gaining understanding of roles, their dependencies and their communication patterns. The teams' collaboration was studied from the perspectives of two socio-technical aspects, i.e. communication and awareness. The communication data were gathered by parsing project documents, to find requirements allocation to roles and their interdependencies, observing the members, questionnaires responses and interviews with members. Awareness data were more rigorous and based on observation and questionnaires. Team members were asked questions through personalised questionnaires regarding their communication with and awareness of other colleagues when working in a team.

6.2. Construction of communication and awareness RCASNs

Based on our observation, questionnaire responses and interview data, we identified 13 reasons for which team members communicated with each other, i.e. bugs discussion, code issues, code reviews, code refactoring, code synchronisation, coordination, management, support issues, sprint planning, quality, user story clarification and user story negotiation. For each of these thirteen communication reasons, communication RCASNs were constructed. In constructing the communication RCASNs, we first identified how many members communicated for certain communication reasons (e.g. "user story negotiation"). Then each member was checked against the other to find their communication for that particular communication reason. In case the answer was "yes", a link was placed between the respective members. The communication links were considered distinct, i.e. if A communicated with B and B communicated with A for user story negotiation, this is considered as two separate communication links. Moreover, the assigned team members were asked to disclose if they had communication with any emergent member or group of members apart from their assigned colleagues for any of the communication reasons. These "emergent members" were also reported in the RCASNs to visualise how much the self-contained agile teams communicate with emergent members in both of the iterations. The emergent members identified in both of the projects were customers, executives, support team and management team.

Likewise, we constructed awareness networks. We divided the concept of awareness into four kinds of awareness, based on the literature, which are: (i) availability awareness, how easy it is for one to reach a person when one needs help for the project (Ehrlich & Chang, 2006); (ii) general awareness, how aware one is of a person's professional background and how his/her skills could help one with his/her work on the project (Ehrlich & Chang, 2006); (iii) current awareness, how aware one is of the current set of tasks that a certain person is working on (Ehrlich & Chang, 2006); and (iv) work status awareness, how aware one is of a colleague's current progress of

work that is related to the project. This fourth type of awareness has been defined by us given the relevance of constant progress report in agile teams. To construct awareness-based networks, data were collected through questionnaires using questions like: (i) "Are you aware of this project member's professional background?" or "Are you aware of how can these project members can help you in your work?" "Is it easy for you to reach this project member when you need help on certain user story?"

6.3. Implementation of the framework

This section evaluates the communication and awareness patterns of agile teams by empirically analysing two projects from two different companies with respect to the proposed framework.

6.3.1. Phase 1 – Conceptualisation

The abbreviations of the underlying concepts, i.e. RCAT and RCASN as defined in Section 4.1 are used hereafter.

6.3.2. Phase 2 – Visualisation

6.3.2.1. Characterisation of actors. Who are involved in the communication and awareness networks? The characterisation of actors showed the details of team members such as their location and average experience. The results showed that in case 1, 4 of 10 members were in Finland and the rest were in Malaysia. However, in case 2, 3 members were in the US and 2 were in Malaysia. The average experience of the team members in case 1 was 3 years and in case 2 it was 1 year. The average number of shared user stories was 4 per iteration for both of the cases.

6.3.2.2. Characterisation of networks. The network characterisation results are described as below.

- **Network visualisation:** How to visualise the ongoing communication and awareness in agile teams? The sociograms created for all of the communication reasons and awareness types helped to illustrate the collaboration among teams. Fig. 4(a) and (b) presents the sociogram for user story clarification communication in cases 1 and 2, and Fig. 4(c) and (d) shows the general awareness network for both cases.
- **Network size:** To answer "How many team members are involved in the communication and awareness networks?" the network size is recorded. The network size determined for both cases and iterations showed that the team members increase only in the case of emergent members' intervention. The major reasons for communication with emergent members include bugs discussion, user story clarification and negotiation with customer, support with support team, and management with management team. The finding somehow corroborates the findings of Damian and colleagues regarding traditional teams that tend to communicate with different teams from what was planned but only in the cases when teams need to communicate with certain emergent members on certain issues (e.g. support team for support issues).
- **Network density:** To answer "How dense are the communication and awareness network?" density measure is applied. The density measures show high values (≥ 0.5) for case 2 as follows: communication changes, coordination, quality, and user story clarification and negotiation. In contrast, for the majority of the remaining networks (in case 1), the density is lower or equal than 0.3, suggesting that only about one-third of the possible communication took place. However, these results do not mean that there was insufficient communication. On manual inspection of the sociograms for the communication reasons that have scores lower or equal to 0.3 (e.g. code review, support, management, and sprint planning in case 2), we noticed that the first

two involve complete ties among most of the developers and the last two between the project manager and one or two senior developers from all cases of both iterations. This suggests that although the scores are low, still there are subgroups strongly connected to each other. For the awareness networks, around half of the networks have a density higher than 0.5. Case 2 shows high density for all networks, which implies that all of the team members were well aware of each other.

- **Network trends:** “How much communication and awareness exists cross and within-roles, cross and within-locations and cross and within-task assignment?” The detailed network ties statistics calculated manually based on the presence of relation between the team members showed that: Within and cross role, more communication happened between cross-roles as compared with the within roles communication. In addition, an increase in the communication ties was noticed for cross and within-roles communication in the second iteration; within and cross sites, more communication was observed cross site as compared with within site communication links, and communication increased by 13% for case 1 and 15% for case 2 in the second iteration. However, awareness ties were large within site and remained the same for both of the iterations. The results reveal that communication among teams increased with project growth. In the case of within and cross-role communication, we had only pairs of developers-quality engineers in case 1 and developers-designers in case 2, as pairs with highest communication. However, most of the within-roles communication happened between developers in both of the iterations. The awareness ties within the same roles increased with time. However, there no significant raise was observed in awareness ties for pairs of distinct roles in the second iteration. The analysis showed that most of the communication took place among

“assigned” members and involvement of emergent members (i.e. customer, executives, support and management team) was noticed to be nominal in both of the iterations. This shows that agile teams work closely with each other and are less dependent on external resources for information seeking. However, awareness of emergent members was considerably less among agile teams. It was noticed that team members had not collaborated with emergent members directly; rather, PM and PO served as intermediaries between team members and emergent members. The general awareness of emergent members increased in iteration 2 as compared with that in iteration 1. Overall, awareness among assigned and emergent roles increased in iteration 2.

6.3.3. Phase 3 – Characterisation

In this section, the results of SNA measures applied for investigating the characteristics of communication and awareness network patterns are presented.

6.3.3.1. Characterisation of communication and awareness RCASNs patterns.

- **Centralisation:** Do agile teams have centralised communication and awareness networks structures? The centralisation results showed that only a small percentage of the communication networks in case 1 (15–23%) and (17–21%) in case 2 have centralisation values close to 1 (≥ 0.60). The analysis reveals that agile teams follow a decentralised communication structure. Moreover, no particular increase or decrease in the centralisation values was observed for both cases and iterations. This proves that the centralisation of communication networks did not increase or decrease with the growth of the project. In awareness RCASNs, the availability and general awareness networks

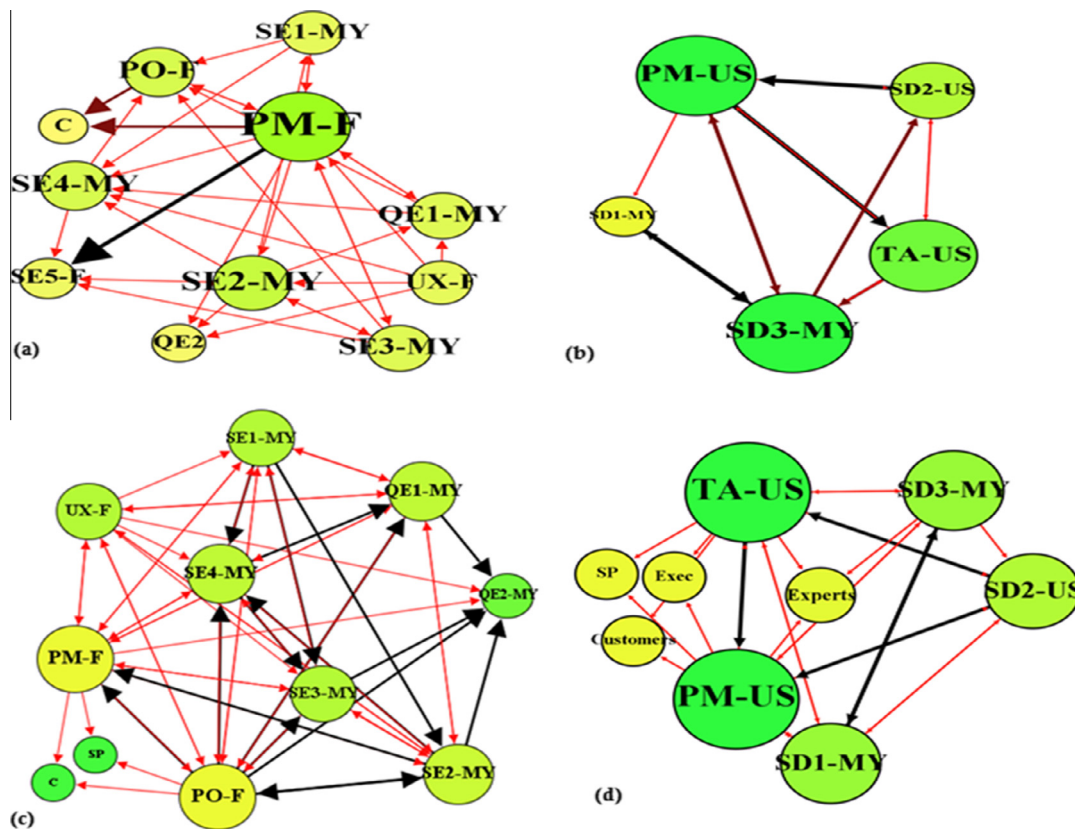


Fig. 4. User story clarification sociograms for (a) case 1 and (b) case 2; general awareness sociograms for (c) case 1 and (d) case 2.

showed somewhat centralised behaviour with centralisation index closer to 1 for out degree centralisation value, for both projects. The mean value for in and out degree centralisation shows centralisation only for out degree centralisation index in iteration 1, and for the rest awareness was dispersed among all the members.

- **Clique:** Does the agile team have cliques? The study of cliques shows very interesting social behaviours of the team members; for instance, PM remained the key member in most of the cliques along with core members like developers and quality engineers in case 1 and developers and designers in case 2. It shows that most of the cliques are organised around the managerial or organisational members of the team and are motivated by the developers.
- **Core periphery:** Who are the core members in the team? The results indicate low core periphery index values for communication networks, which show that networks have more decentralised structures. However, the managerial members, i.e. PM and PO appeared in almost all the networks as core members except for code issues, in iteration 1. PM preceded the PO in this because PM was the one in close contact with the team, PO and emergent members simultaneously. In case 1, UX was always found in the periphery group, which shows that he was loosely connected with other team members. As explained earlier, UX was not a dedicated member of the team; hence, his involvement with this project and team was minimal. Moreover, the core members also exhibit high awareness of others.

6.3.3.2. Characterisation of information flow patterns. To identify the patterns of knowledge exchange in the agile team, several measures to determine the information flow are used, such as component, degree centrality, reachability, and cut points. The results are discussed with respect to the research questions below:

- **Centrality:** Do agile teams have some actors playing a central part in information flow? The nodes with recurrent high out degree values are PM and SD2 in case 1 and PM and SD3 in case 2. The developers remained active and prominent in sending information regarding code issues, refactoring, reviews, synchronisation and coordination of activities. On the contrary, no such recurrent pattern can be seen in the case of in degree. The least values of in degree are identified for code issues, code reviews and code refactoring. The reason behind this is participation of only core members, i.e. developers in code related communication. Therefore, it can be concluded that PM was the powerful role in this team and was responsible for imparting most of the information to the rest of the team members.

Table 4

Performance estimation factors.

Performance measures	Case1		Case2	
	It 1	It 2	It 1	It 2
No. of tasks allocated per iteration	8	10	7	5
No. of tasks completed	6	6	5	4
No. of tasks not completed	2	4	2	1
Total story points allocated per iteration	57	43	61	45
Story points per iteration (velocity)	49	33	49	37

However, for code related communication, developers were powerful actors.

- **Cut point:** Do agile teams have such member(s) who play a pivotal part in information exchange? PM and SD2 were the cut points in case 1 and PM and SD3 in case 2, which if removed would result in communication lapse in most of the communication networks, in both of the iterations. SD2 was the senior member with experience level of 5–7 years working at company Alpha, and SD3 was the team leader with 6 years of work experience. Both were the focal point of most of the discussions regarding code or otherwise along with PM. This way, the managers can see the focal team members and foresee the information loss and communication lapse in case of their absence or resignation.
- **Clustering coefficients:** Do agile teams have the tendency to cluster with the increase in communication? Overall, low clustering tendency was recorded in communication networks in both of the iterations for case 1. Case 2 has slight tendency towards clustering as compared with case 1, i.e. 38% and 46% of the communication networks had cliques for both of the iterations respectively. Therefore, overall we can say that communication networks did not exhibit clustering tendency. The clustering coefficient values calculated for awareness networks were contrary to communication networks. The awareness networks showed high values of clustering coefficients, which is a confirmation of more cliques. However, case 2 showed less awareness clustering as compared with case 1. Therefore, it can be concluded that number of cliques in awareness networks increased with time awareness, which means that team members gained mutual acquaintance of each other.
- **Small world network structures:** Does the agile team show small world network structure? The small worldliness was calculated for communication networks of case 1 and case 2, for both iterations. The results are shown in Fig. 5(a) and (b). The results revealed that the small worldliness of the communication patterns increased in iteration 2. However, overall small worldliness index remained low for most of the communication networks, which shows that the team did not show highly centralised structures.

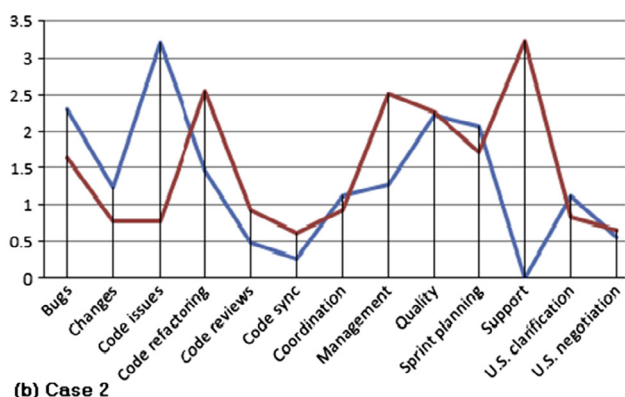
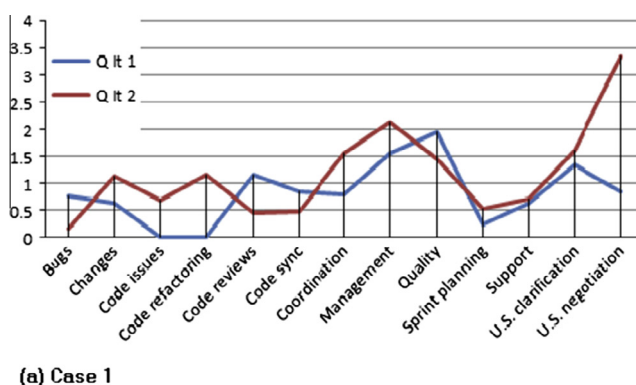


Fig. 5. Small world network structure values for communication RCASNs for (a) case 1 and (b) case 2.

6.3.4. Phase 4 – Impact analysis

How do the small worldliness and centralisation affect iteration performance? The performance of the teams was calculated by taking into account several measures as shown in Table 4. The multiple regression method is used in order to predict iteration performance.

The results revealed that small worldliness of the networks strongly affects the iteration performance in a negative direction ($\beta = -0.133, t = 0.357$ at $p < 0.05$). This corroborates the findings of Cataldo and Ehrlich (2011) that small world network structures negatively affect iteration performance of distributed agile teams results. However, the experimental conditions, team size, communication data and data collection methods employed by the prior research study differ from this research design and settings. Moreover, Cataldo and Ehrlich (2011) determined iteration performance using number of leftover tasks per iteration and did not consider velocity or story points in their study.

When tested for centralisation, the results revealed that out degree centralisation has a strong and negative effect on the iteration performance ($\beta = -0.459, t = -0.972$ at $p < 0.05$). However, in degree centralisation has slight positive effect on the team performance ($\beta = 0.228, t = 0.357$ at $p < 0.05$). Therefore, it can be concluded that in degree centralisation results corroborate the previous results stating that low centralisation indicates better team performance (Sparrowe, Liden, Wayne, & Kraimer, 2001). However, we propose that low out degree centralisation has negative effect on iteration performance.

7. Concluding remarks

Requirements engineering and agile methods, both have collaboration oriented nature. However, socio-technical aspects of collaboration need to be explored while dealing with requirements in agile teams. Therefore, a framework is proposed in this research to study RDC among agile teams through the lens of communication and awareness. The proposed framework is structurally validated to gather feedback on its constructs through semi-structured interviews with experts in the field. The suggestions gathered from the experts are then implemented. The framework is then practically demonstrated through a multiple case study performed at two software development organisations that have adopted agile methods, with teams distributed in Asia, Europe and America. The framework helped in studying the collaboration among teams and the findings revealed the characteristics of the networks such as: overall communication and awareness increased in iteration 2; and project managers, owner and the experienced software developer were the core members. Likewise, the examination of networks' characteristics revealed that most of the networks showed decentralised behaviours. Regarding the information flow patterns of agile teams, it is recorded that the small worldliness increased in iteration 2 as compared with that in iteration 1. Moreover, it was found that communication and awareness are interdependent, and distance affects communication and awareness among team members. It is revealed that the agile teams' performance is negatively affected by small worldliness and centralised network structures. Finally, we draw implications and future directions for the researchers and industry practitioners. The proposed framework provides the managers and researchers with an equal opportunity to study their team's collaboration during the software development life cycle and take measures to improve their performance. Moreover, the research practitioners can use this framework as a guideline to study more social aspects in different team settings to produce more empirical results.

8. Implications for research and industry practitioners

The results of this study can be used by industry practitioners to observe the insights of team's collaboration in their organisations and improve their collaboration practices by diagnosing communication gaps and missing links. This framework is a formative evaluation mechanism for researchers so that they can identify potential new research areas and as an assessment strategy for potential industry users to understand collaboration patterns of their teams and relate them to their performance. Some of the implications for industry practitioners are:

1. Keeping track of team's collaboration: Agile teams are meant to collaborate with each other rigorously. Therefore, this framework can aid the managers to see their team's collaboration practices (i.e. frequent communication with collocated than remote), trends (i.e. likelihood of members' communication with and awareness of others) and flaws (i.e. missing communication or lack of awareness) at any instance during the project life cycle. The managers can keep a constant check on reduced or increased collaboration among members through repeated checks. As a result, the manager can take necessary action to orient the teams in a better way to incorporate more collaboration among teams for better performance.
2. Better team-orientation decisions: The managers can make use of their team's awareness level knowledge to group them for future teams. The team members who are well aware of each other can communicate frequently, and this can directly affect the outcome. Therefore, the managers get plenty of useful information to make such important decisions in the future.
3. Improve collaboration infrastructure among distributed teams: Managers can invest in having well-defined infrastructure in place to allow team members to contact their remote colleagues. The collaboration among remote members is highly important for an agile team to carry out work together in a two-week iteration. Therefore, communication lapses among distributed members can be dealt with by making use of this mechanism.
4. Comprehensive tool design: The tool designers can make use of this study to develop a comprehensive tool which collects data, suggests SNA measures to be applied to it to get appropriate results based on the manager's queries. The impact of certain features on each other and the team's overall performance can also be incorporated.

Some of the implications for research practitioners are:

Further researches can be carried out to provide more empirical results using different configurations such as larger teams to check if the project team size affects collaboration behaviour, different project types, e.g. outsourced projects and open source projects to identify whether the nature of the project has any influence on collaboration patterns, and different business domain to find out to the extent to which the domain knowledge affects team collaboration. Moreover, other socio-technical aspects like teamwork, trust and organisational structure should also be studied within agile teams to find out their respective collaboration patterns and their due effects on team performance. The performance measures can be extended to other possible measures used in industry such as throughput.

9. Directions for future research

This research provides directions for future research such as:

1. Case studies with other configurations such as larger teams, larger number of requirements dependencies, different business background, and higher physical distribution to further the knowledge on the effect of distance on RDC, can be some of the potential cases to be investigated for adding value to the results.
2. The empirical results can be extended by replicating the study for other agile software development methods such as Kanban and Scrum-ban. The difference in results can lead to further insights into the agile team collaboration patterns and their effects on team performance.
3. In terms of performance analysis, more performance metrics other than burn down user stories and velocity need to be incorporated in order to find the detailed view of the collaboration effects on each of them. Moreover, the quality of outcome and user story needs to be considered in addition to the iteration performance.

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Appendix A

See Table A.1.

Table A.1

Interview guideline for conducting interviews with industry practitioners.

Demographics					
Gender					
Role					
Work experience in present role					
			Less than 1 year		
			Less than 3 years		
			Less than 5 years		
			Less than 7 years		
			Above 7 years		
Your company is			Small, medium or large sized		
1. The framework has following phases. Do you think all of them are necessary?	Strongly agree	Agree	Neutral	Disagree	Strongly disagree
(a) The conceptualization phase, (b) the visualization phase, (c) the characterization of communication and awareness among agile teams, and (d) the impact analysis phase					
2. The framework is understandable step by step for the following phases:					
(a) The conceptualization phase, (b) the visualization phase, (c) the characterization of communication					

and awareness among agile teams, (d) the impact analysis phase					
3. How do you rate the complexity level of this framework as:	Very easy	Easy	Fair	Difficult	Very difficult
4. How do you think the framework has potential to be practically implemented	Strongly agree	Agree	Neutral	Disagree	Strongly disagree
5. Do you look forward to use this framework for studying collaboration and performance analysis of your team in future?					
6. How do you think the results can help you to see the insights of your team's collaboration practices?	Very helpful	Helpful	Neutral	Useless	Very useless
7. How do you think the impact analysis can help you to assess the performance of your team in the light of their collaboration practices?					
8. Any suggestions for improvement?					

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